CA-3 | Agentic AI Proof of Concept

Model Drift Watchdog

An Agentic AI System That Detects, Decides, Validates & Acts When ML Models Go Stale

Domain: MLOps · Credit Risk in Digital Lending

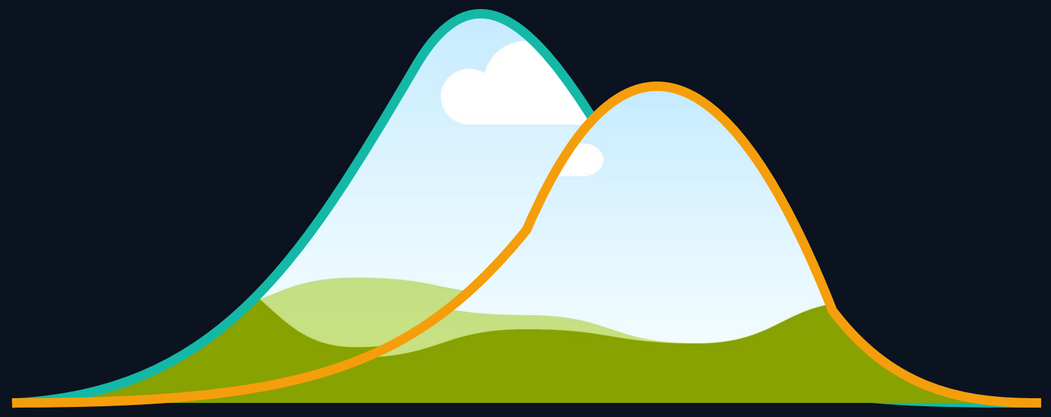
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Presenter: Harsh

ML Models Fail Silently

- Models trained on past data; real-world data changes
- Incomes rise, new customer types appear, pipelines break
- Model keeps running but makes worse decisions
- Risky loans approved; good customers rejected in lending
- Loan defaults known months later → losses discovered too late



Data Drift = the input data in production no longer looks like the training data

Presenter: Harsh

Why MLOps in Lending?

High Financial Stakes

Every wrong prediction is a bad loan or a lost customer

Label Delay

True outcomes arrive months later; input monitoring is the only early signal

Regulatory Risk

Lenders must explain and audit automated credit decisions

Monitoring is a real, everyday pain point for ML teams in industry.

From Manual Checks to Agentic Monitoring

Presenter: Harsh

	Today (Manual) ×	With Model Drift Watchdog ✓
Frequency	× Monthly / ad-hoc notebook checks	✓ Every new data batch, automatically
Detection	× Noticed after losses appear	✓ Caught early from input-data drift
Root cause	× Engineers investigate by hand	✓ Agents separate real drift from broken pipelines
Decision	× Gut feeling, inconsistent	✓ Policy-based, validated by a Critic agent
Retraining	× Manual and risky	✓ Old vs new model comparison + human approval
Reporting	× None or informal	✓ Auto-generated report and alert

Result: faster detection · consistent decisions · fewer false alarms · full audit trail

6 Agents + 1 Human, One Mission

Presenter: Harsh

PLANNER AGENT

Planner

Builds a task plan from the data; re-plans if data is bad

DATA PROFILER

Research

Checks schema, missing values, sample size, statistics

DRIFT DETECTOR

Executor

Runs KS, Chi-square & PSI tests on every feature

DECISION AGENT

Business Automation

Applies policy: No Action / Alert / Retrain / Fix Pipeline

CRITIC AGENT

Critic

Validates evidence and decision; triggers retries or escalation

REPORTER / ACTION

Executor

Writes report & alert; retrains and compares models

HUMAN APPROVER ◉ Final sign-off before any new model is deployed

System Architecture

Presenter: Harsh

DATA
reference.csv (training data)
production batch (per scenario)
champion model

Streamlit UI
Pick scenario · Run watchdog · View trace · Approve

request + new data batch

Planner Agent — Labels present? Feature types? → Task plan

Data Profiler — Schema · Missing values · Sample size

Missing >10% / schema error
RE-PLAN → Fix Pipeline

data OK

Drift Detector — KS · Chi² · PSI · Prediction drift · Accuracy (if labels)

Decision Agent
No Action · Alert · Retrain · Fix Pipeline

Critic Agent ◇ — Retry (max 2) · Still failing → Escalate to Human

Reporter / Action Agent — Report · Alert · Retrain challenger · Champion vs Challenger (AUC)

Human Approval — Deploy new model / Close incident

TOOLS
pandas · NumPy
SciPy (KS, Chi²) · PSI
scikit-learn · Plotly
LangGraph
Ollama / Mock LLM

Orchestrated with LangGraph: shared state + conditional edges for re-planning, retries and escalation

How the Agents Decide — and Double-Check

Presenter: Divyansh

Decision Policy

PSI < 0.10 Stable | 0.10–0.25 Moderate | > 0.25 Major | p-value < 0.05 = Drift

FIX PIPELINE Missing values > 10% or schema mismatch

RETRAIN Major drift on a critical feature, prediction PSI > 0.25, or accuracy drop > 5 pts

ALERT Moderate drift, or major drift on a non-critical feature

NO ACTION Everything stable

Critical features: credit_score, monthly_income, loan_amount, employment_type

Critic Checks

◇ Sample size < 500 → Profiler uses a larger window

◇ KS & PSI disagree → Detector re-checks with bootstrap

◇ Decision breaks policy → Decision agent revises

◇ Still failing after 2 retries → Escalate to human

◇ New model deployed only if AUC improves ≥ 0.01 AND human approves

Design principle: Decisions come from statistics and rules (auditable & repeatable). The LLM only explains them in plain English.

4 Synthetic Scenarios We Test

Presenter: Divyansh

Synthetic loan data: 5,000 training rows + 1,000-row production batch · Features: age, monthly income, loan amount, credit score, employment type, city tier · Target: default

Normal Week

Injected:
Nothing changed

Expected:
✅ No Action

Gradual Drift

Injected:
Incomes up ~20%, city tier shifts

Expected:
⚠ Alert

Sudden Drift

Injected:
Influx of gig workers, lower credit scores

Expected:
🔁 Retrain → Human approves

Pipeline Bug

Injected:
40% credit scores missing

Expected:
🛠 Fix Pipeline (NOT retrain)

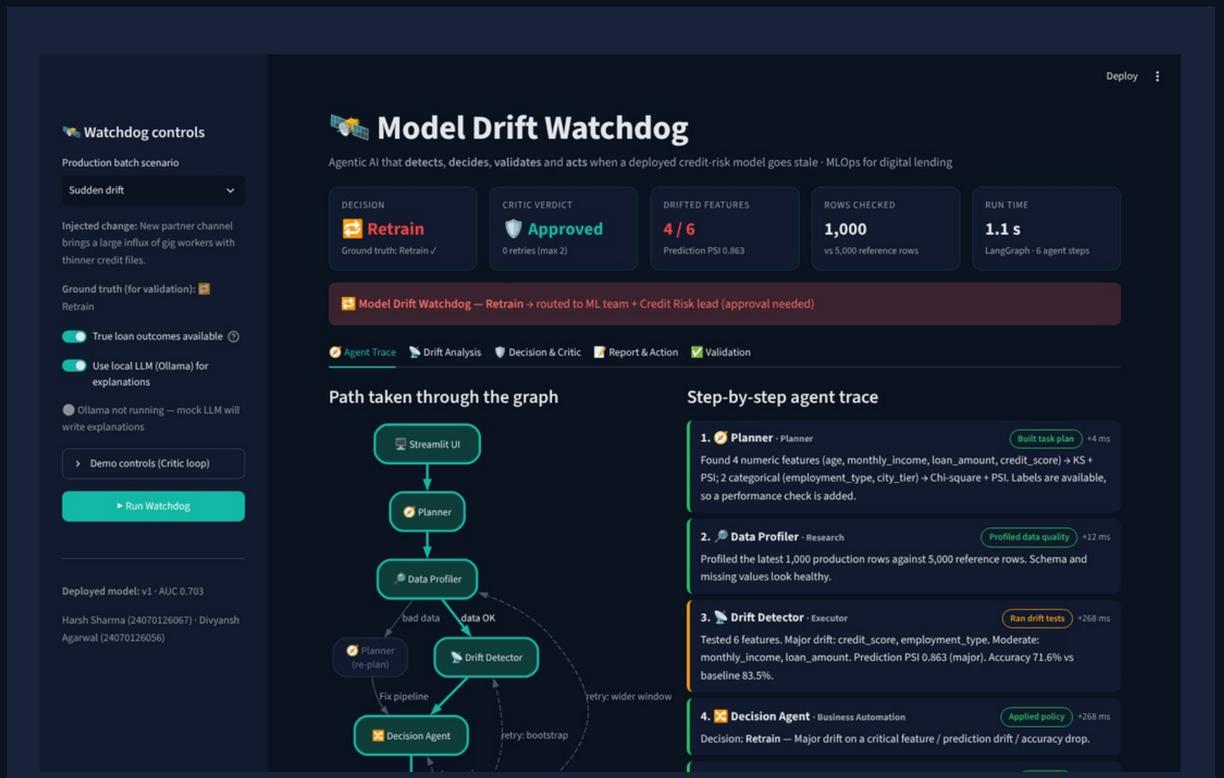
Scenario 4 is the key test: a naive system would retrain on broken data — our agents don't.

Live Demo — Streamlit PoC

Presenter: Divyansh

1 Sidebar
scenario picker · labels toggle · Ollama on/off · Run

2 Agent Trace
step-by-step timeline of every agent, including retries



3 Drift Analysis
reference vs production charts · PSI bars · drift table

4 Decision & Critic
triggered rule + pass/fail checklist

5 Report & Action
plain-English report · champion vs challenger · Approve / Reject

Demo flow: Normal Week → Sudden Drift → Pipeline Bug

How We Validate the Output

Presenter: Divyansh

5 Layers of Validation

1 Ground truth: we know what drift was injected

2 Detection check: drifted features vs flagged features

3 Critic checks: sample size, test agreement, policy

4 Model check: challenger must beat champion on fresh data

5 Human check: nothing deploys without approval

Scenario | Expected | Agent Decision | Match

Normal Week | No Action | No Action | ✅

Gradual Drift | Alert | Alert | ✅

Sudden Drift | Retrain | Retrain | ✅

Pipeline Bug | Fix Pipeline | Fix Pipeline | ✅

8/8 correct (incl. label delay) · Pipeline bug NOT retrained
Retrain: +0.026 AUC on fresh data · 7/7 robustness tests

Limitations & What's Next

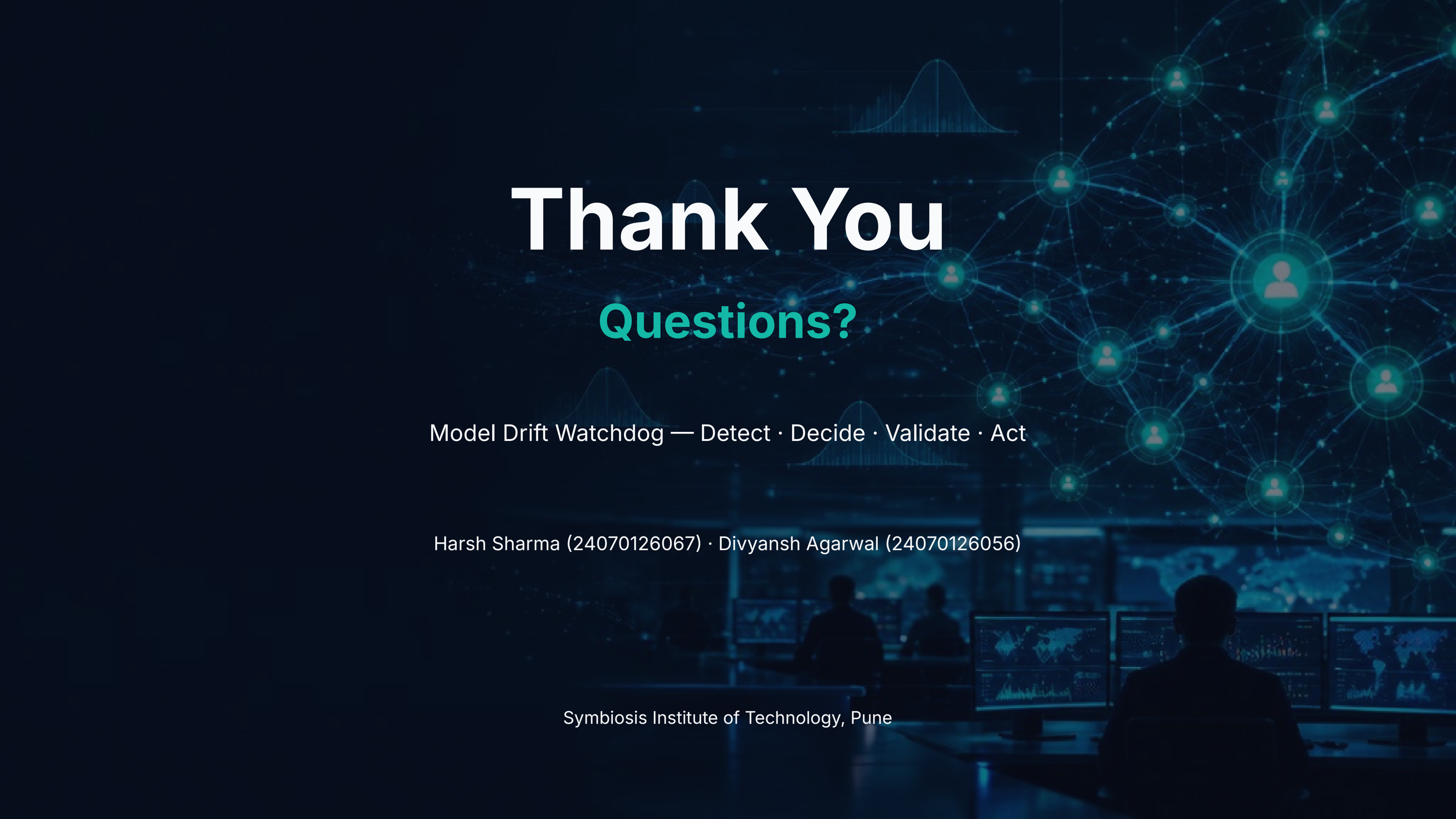
Presenter: Divyansh

Limitations

- Synthetic data only
- Per-feature tests can miss combined (multivariate) drift
- Fixed thresholds, not learned
- Label delay not simulated
- Mock LLM explanations are template-based

Future Scope

- ↗ Scheduled / real-time streaming monitoring
- ↗ Multivariate drift detection (classifier-based)
- ↗ MLflow model registry integration
- ↗ Slack / email alerts via MCP
- ↗ Canary rollout of new models & adaptive thresholds



Thank You

Questions?

Model Drift Watchdog — Detect · Decide · Validate · Act

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